

How BettaTool was trained

One-page handout for the thesis defense. Code: backend/training/. Settings: configs/base.yaml, configs/hrnet_w32.yaml.

The idea

Show a photo, let the model **guess, score** how wrong it is (the loss), work out which weights to **blame, nudge** them, repeat. The model gives two answers per keypoint: **where** it is (position) and **how unsure** it is (doubt, a 2x2 covariance Σ with σ_x, σ_y, ρ).

Stage 1 (40 epochs): learn position only (heatmap KL + L1). **Stage 2 (80 epochs):** also learn doubt (β -NLL), turned up 0→1 over 10 epochs. Epoch = one pass over the ≈1,160 training photos. Batch = 8 photos; a weight step happens every 2 batches (16 photos).

Total loss (train_hrnet.py:217-226)

```
total = 1.0·KL + 1.0·L1 +  $\lambda_{nll}$ · $\beta$ NLL then ÷ 2 (accumulation)
 $\lambda_{nll}$  = min(1, epoch ÷ 10) in stage 2, else 0 · other weights = 1.0
```

1. Heatmap KL: does the glow map look like the answer?

```
KL = sum over pixels of  $T \times (\log T - \log P)$ 
T = target glow map ÷ its total (sums to 1). Target glow =  $\exp(-(dx^2+dy^2) \div (2s^2))$  around the label,  $s = 2$  heatmap px (fin tips: 4 px across the fin). P = predicted glow map (spatial softmax, sums to 1). File: losses/heatmap_kl.py
Example: prediction off by 0, 1, 2, 4 heatmap px gives KL = 0, 0.125, 0.50, 2.00. Big misses cost much more.
```

2. L1: is the final position close?

```
L1 =  $|x_{\text{hat}} - x| + |y_{\text{hat}} - y|$  (soft-argmax, ×4 to photo px)
 $x_{\text{hat}}$  = sum of  $x \times P(x,y)$ , the glow-weighted average position. Averaged over visible keypoints (ambiguous = ½ weight), then over the batch. File: train_hrnet.py:204
Example: label (160, 224), guess (166, 220) gives 6 + 4 = 10.
```

3. Doubt loss (β -NLL): is your stated doubt honest?

```
NLL =  $0.5 \cdot \log \det(\Sigma) + 0.5 \cdot r^T \cdot \text{inv}(\Sigma) \cdot r$  ( $r = \text{label} - \mu$ )
 $\beta$ NLL = NLL ×  $(0.5 \cdot \text{trace } \Sigma)^\beta$ ,  $\beta = 0.5$  (weight itself not trained)
 $\Sigma = [[\sigma_x^2, \rho\sigma_x\sigma_y], [\rho\sigma_x\sigma_y, \sigma_y^2]]$ ,  $\sigma$  clamped 0.5–64 px,  $|\rho| \leq 0.95$ . File: losses/gaussian_nll.py. The first term punishes being vague; the r-term punishes being wrong while sure, so the best score is honest doubt.
Example (10 px error, plain NLL):  $\sigma=2$  gives 13.9,  $\sigma=5$  gives 5.2,  $\sigma=7$  gives 4.9 (best),  $\sigma=10$  gives 5.1,  $\sigma=14$  gives 5.5.  $nll\_detach\_mu = \text{true}$ : the NLL trains the doubt, not the position.
```

Messy-label tricks (fin tips are noisy)

- **Wider fin target:** across-fin $\sigma \times 2$ (5 fin keypoints).
- **Ambiguous labels count ½** (KL and L1).
- **Teacher blend** from stage-2 epoch 15: target = $(1 - \alpha) \cdot \text{label} + \alpha \cdot \text{teacher}$; $\alpha = 0.25$ on fin keypoints, +0.5 if ambiguous (max 1). Example: label 100, teacher 110, $\alpha = 0.25$ gives target **102.5**. L1 only.
- **Teacher = EMA** of the model: $\text{ema} = 0.998 \cdot \text{ema} + 0.002 \cdot \text{model}$ (the same model, slowly averaged).
- **Trim:** from epoch 15, ignore the worst 10% of fin-tip L1 residuals in each batch (L1 only).

Control and checkpoints

Learning rate = nudge size. Warm-up 3 epochs from 0.1×, then cosine decay to about 0; restarts at stage 2. Heads peak 0.001, backbone 0.0001 (pre-trained, gentler). **Safety:** gradient clip 1.0, weight decay 1e-4, mixed precision. **Validation** each epoch (no augmentation). **Early stopping** on validation NLL, patience 40. **Deployed file:** best_nll.pt = stage 2, epoch 20, val NLL 3.97 (position error alone would wrongly pick epoch 0).

Important configs

Setting	Value
Backbone / input	HRNet-W32 (ImageNet), 384×384 → heatmap 96×96 (stride 4), 13 keypoints
Split	70 / 15 / 15 (on disk 1161 / 246 / 247); seed 42
Epochs	stage 1 = 40, stage 2 = 80 (base.yaml says 110; overridden)
Batch	8 × accumulation 2 = 16
Optimizer	AdamW; lr heads 1e-3, backbone 1e-4; weight decay 1e-4
Schedule	warm-up 3 epochs + cosine; restart at stage 2
Grad clip / AMP	1.0 / on
EMA	decay 0.998 (about 7-epoch horizon)
Loss weights	$\lambda_{\text{heatmap}} = \lambda_{\text{L1}} = \lambda_{\text{nll}} = 1.0$; NLL ramp 10 epochs
β-NLL	$\beta = 0.5$; $nll_detach_mu = \text{true}$
Covariance	σ 0.5–64 px; $ \rho \leq 0.95$
Heatmap target	σ 2 px; fin tips across ×2.0
Noisy labels	ambiguous weight 0.5; α fin 0.25, ambiguous 0.5; teacher + trim from epoch 15; trim 10%
Early stopping	val NLL, patience 40, from stage 2
Augmentation (train only)	flip 0.5; rotate ±15°; scale 0.85–1.15; shift ±5%; colour jitter (p 0.8); blur 0.2; JPEG 0.2; random holes 0.3

Where each piece lives

Training loop, scores, trimming, teacher: training/train_hrnet.py. Heatmap targets: training/targets.py. Losses: training/losses/. EMA: training/utils/ema.py. Augmentation: training/transforms.py. Model heads and soft-argmax: app/perception/hrnet.py, heatmap.py. The learned σ becomes the gate's predicted error sigma-hat, compared with the TSI.